VULTR AGENT RUSH · CHALLENGE 02 · FINANCE & ACCOUNTING

# Ledgerline

## A bank reconciliation agent that does not guess.

The code does the math, the model writes the words, and every number it writes is checked.

Upload a bank statement and a ledger (CSV). Get matched lines, exceptions with evidence, and reminder drafts a person approves.

Runs on Vultr Cloud Compute + Vultr Serverless Inference · live demo and code linked in the README

MIT licensed

PROBLEM

# Month-end matching is manual, and mistakes are expensive

- Someone in finance puts the bank statement next to the ledger and ticks off lines one by one.
- Exports never look the same: `1.234,56` vs `1,234.56`, day-first vs month-first dates, Debit/Credit vs one signed column, preamble lines, other languages.
- The hard part is the exceptions: a fee deducted from a wire, one transfer paying two invoices, a double charge, two invoices with the same amount.
- A chat assistant can do the arithmetic, but it can also make up a number. In bookkeeping, one invented number costs more than the time it saved.
- **Who has this problem:** small finance teams closing the month in spreadsheets, and bookkeeping firms doing it for many clients with many bank formats.

SOLUTION

# An agent with a fixed rule about who does what

### Code: every number

Column detection, date format, sign convention, matching rules R1 to R4 (reference, amount + date, split payments, short payments), differences, totals per currency, days overdue. All in `Decimal`, each result tagged with its rule.

### Model: the words

Explains each exception and drafts payment reminders, on Vultr Serverless Inference. A checker removes any sentence whose numbers are not in the evidence, and the UI shows what it removed.

### Person: the decisions

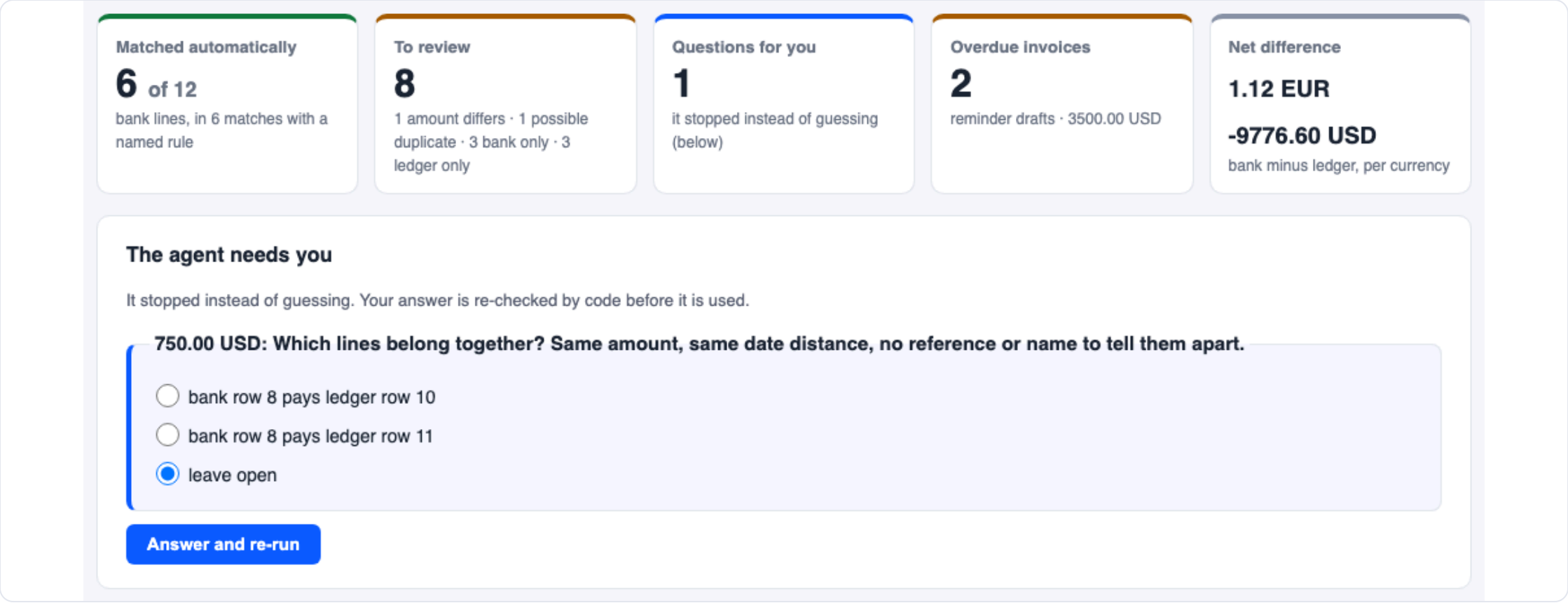
If two candidates are equally likely, a column is unclear or a date could be read both ways, the agent stops and asks. Reminders go out only after a person approves them, and each approval is logged.

### Nothing hidden

Unreadable files raise an error instead of an empty result. Unreadable rows show up as exceptions. If the model is down, template text appears with the reason.

PRODUCT

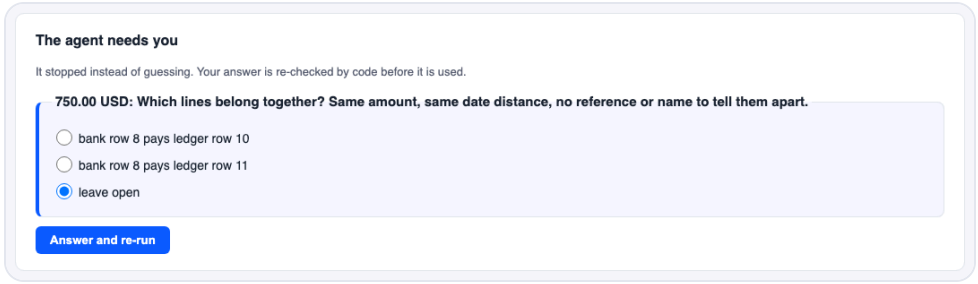
# One click, and the reviewer sees only what needs a person



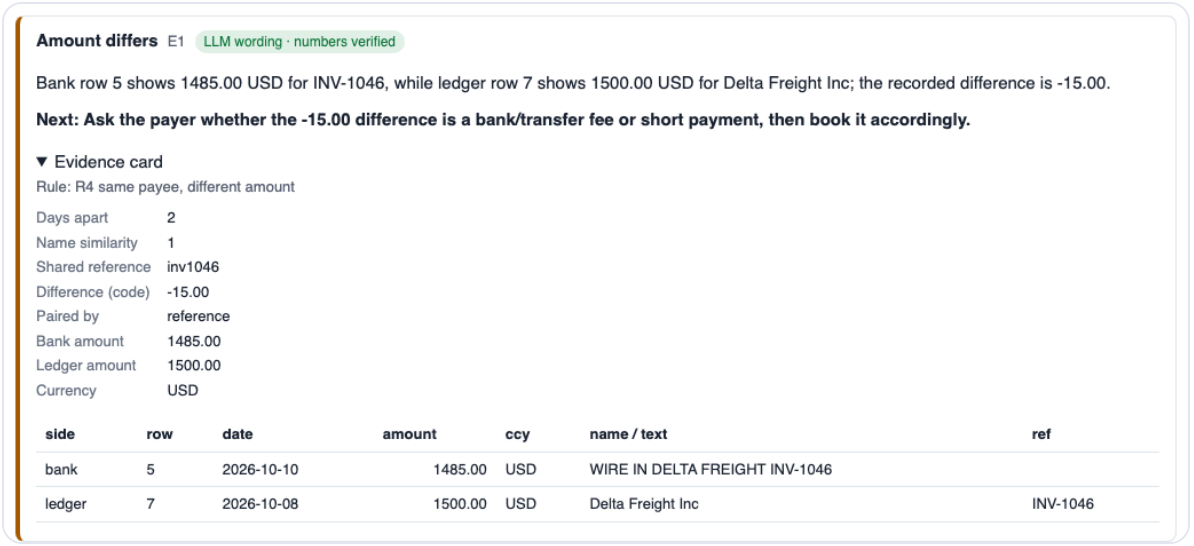

Summary cards counted by code: lines matched by a named rule, exceptions to review, questions it stopped on, overdue invoices and the net difference per currency. Results download as a CSV. Three sample sets (US, German, UK exports) run from the first screen.

DEMO

# It asks, then explains with evidence



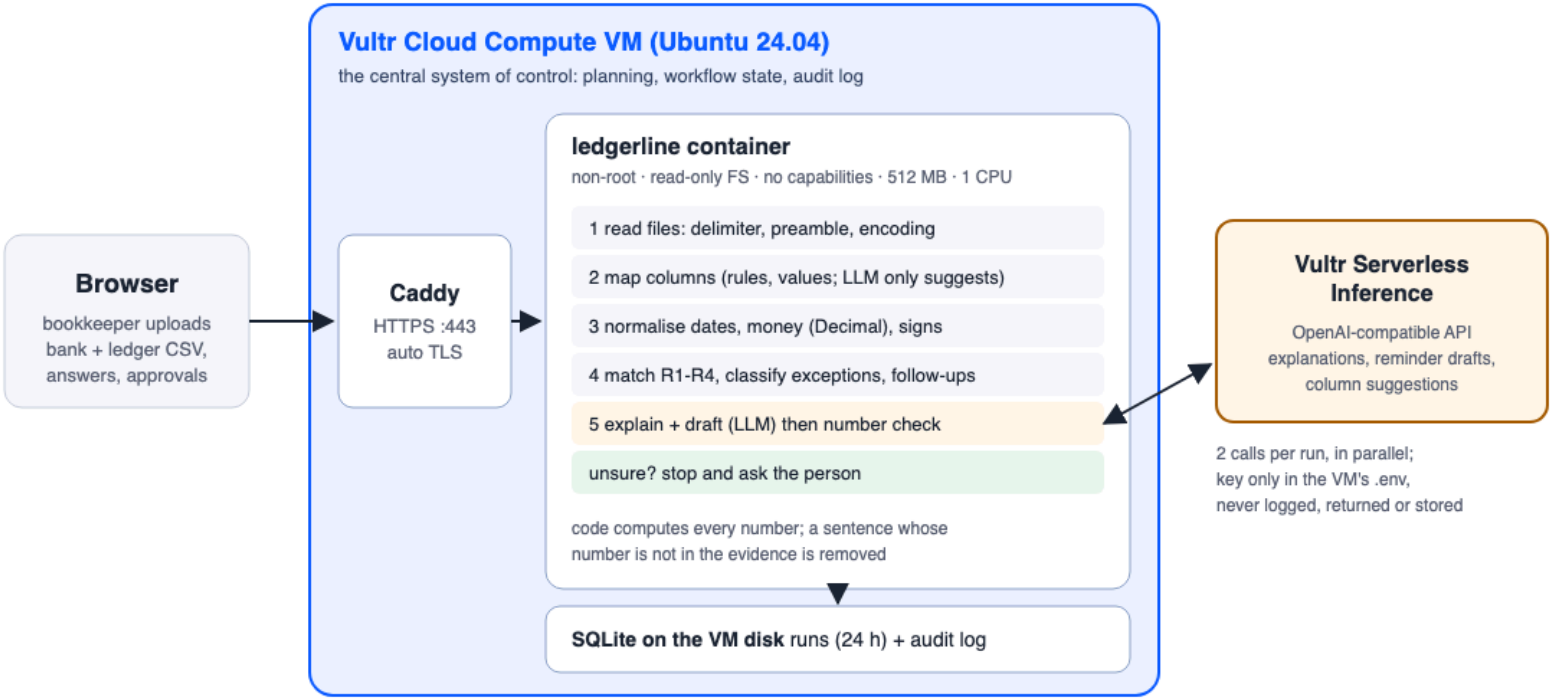

A $750 deposit with no name, and two $750 invoices from the same day. It asks instead of flipping a coin.




Delta Freight paid $15.00 less. Code computed the difference and the model wrote the sentence; every number in it is verified.

ARCHITECTURE ON VULTR

# One VM is the system of control




- **Why Vultr:** the VM runs every step, the run state and the approval audit log; Serverless Inference writes the words. No other cloud service, and with the model off no data leaves the VM.
- Deployed with one script: `deploy/vultr-setup.sh` (Docker + Caddy HTTPS). Live: `https://64-176-236-185.sslip.io`, model `deepseek-v4.1-flash-1017` on Vultr Serverless Inference.

EVALUATION

# Tested on 20 file pairs someone else wrote without seeing the code

| | First sight | After fixes | Live on Vultr (public URL, Vultr Serverless Inference) |
|---|---|---|---|
| Set 1 (10 cases) | **8 / 10** | 10 / 10 | 10 / 10 · median 2.3 s · max 3.1 s |
| Holdout set (10 cases, written after the first fixes) | **8 / 10** | 10 / 10 | 10 / 10 · median 2.2 s · max 3.0 s |

"Exact" means every match group, exception, difference and question equals the expected answer. The four first-sight failures (headers it did not know, such as `Txn Desc`, `Payments` and `N° pièce`; a year read as an invoice number; two real identical payments flagged as a duplicate; `$` assumed to be USD) and their fixes are listed in `eval/RESULTS.md`. An independent code review found 15 more defects, and each is now a regression test. In total, 72 tests cover these cases, including the model inventing a number and the model being down.

BUSINESS VALUE

# Who pays, and why

## The work exists today (sourced)

In the US alone, bookkeeping, accounting and auditing clerks held about 1.5 million jobs in 2025, with a median wage of $50,670 (May 2025). Their listed duties include "Reconcile or note and report any differences they find in the records." Employment is projected to decline 6% from 2025 to 2035.

Source: U.S. Bureau of Labor Statistics, Occupational Outlook Handbook, Bookkeeping, Accounting, and Auditing Clerks (read 2026-11-06).

- **Small finance teams** closing the month with spreadsheets: they review only the exceptions, each with evidence and a next step.
- **Bookkeeping firms** running many client entities: one tool for every bank format, plus an audit trail of who approved what.

## Revenue model (our hypothesis, not yet tested)

A monthly subscription for each company, priced by the number of bank accounts. Firms get a multi-client plan. Inference cost stays small because the model writes short text and never sees whole ledgers.

## Why now

Hosted open models make the wording step cheap. The missing piece is an agent that finance teams can trust with numbers, and that is the part we built.

NEXT

# Roadmap

- Bank feeds and PDF statements in addition to CSV. Several accounts and inter-account transfers in one run.
- Learn each client's recurring patterns, such as the usual wire fee, but only as a suggestion the user confirms.
- Send approved reminders through the customer's own mail, and write matches back to QuickBooks/Xero.
- Measure the claim we have not measured yet: time a real month-end close with and without Ledgerline at a small bookkeeping firm.
- Tighter number checks that tie each number to the field it came from, not just "it appears in the evidence".

Code: github.com/dormkim/ledgerline (MIT) · Live demo: https://64-176-236-185.sslip.io · Evaluation data and results are in the repo.